

BINARY SEARCH & complexity

The idea DEFINITION

Binary search — repeatedly halve a sorted array: compare the target with the middle element, and discard the half that cannot contain it. Each comparison throws away **half of the remaining candidates**.

Precondition array is sorted

Time $O(\log n)$

Space $O(1)$ iterative

The loop invariant

Everything the proof of correctness rests on:

- If the target is in the array, it lies in $a[lo..hi]$.
- Every iteration **strictly shrinks** the interval, so the loop terminates.
- The interval is empty when $lo > hi$ — the target is absent.

The algorithm

$lo \leftarrow 0, hi \leftarrow n - 1$
the interval starts as the whole array

→ Invariant holds trivially at entry

while $lo \leq hi$
 $mid \leftarrow lo + (hi - lo) / 2$

→ Written this way to avoid integer overflow on $lo + hi$

$a[mid] = target \rightarrow \text{RETURN } mid$

→ Found — best case, one comparison

$a[mid] < target \rightarrow lo \leftarrow mid + 1$
 $a[mid] > target \rightarrow hi \leftarrow mid - 1$
discard the half that cannot contain the target — mid included

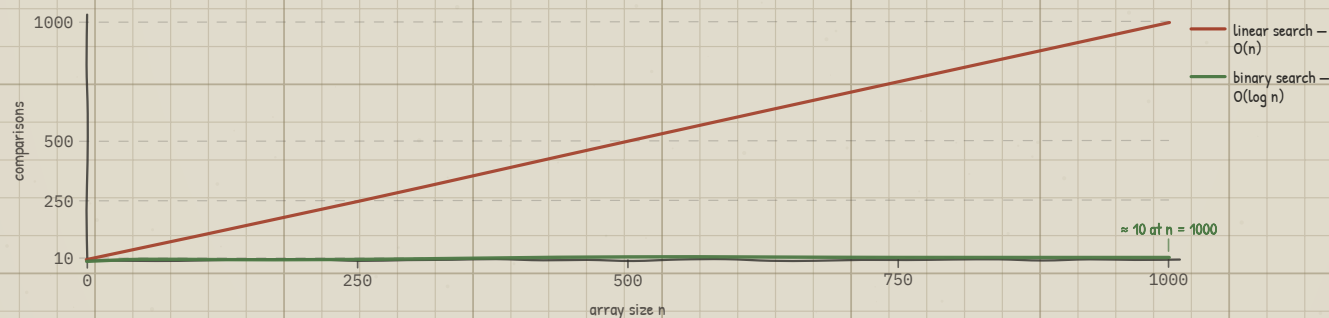
→ Excluding mid is what guarantees the interval shrinks

loop ends $\rightarrow \text{RETURN not found}$

→ Interval empty; the invariant proves absence

Each pass halves the search interval — so the number of passes is the number of times n can be halved.

Why it matters at scale



Linear search reads the whole array; binary search reads a number of elements that barely moves as n grows.

How many comparisons? WORKED EXAMPLE

GIVEN A sorted array of $n = 1\,000\,000$ elements

FORMULA comparisons $\approx \lfloor \log_2 n \rfloor + 1$

SUBSTITUTION $\log_2(1\,000\,000) \approx 19.93$

CALCULATION $\lfloor 19.93 \rfloor + 1$

20 comparisons, worst case

A linear scan of the same array needs up to 1 000 000 — a factor of 50 000.

Cost of the operations around it

Operation	Sorted array	Hash table
Search	$O(\log n)$	$O(1)$ avg
Insert	$O(n)$	$O(1)$ avg
Delete	$O(n)$	$O(1)$ avg
Min / max	$O(1)$	$O(n)$
In-order scan	$O(n)$	$O(n \log n)$

Where it goes wrong COMMON MISTAKE

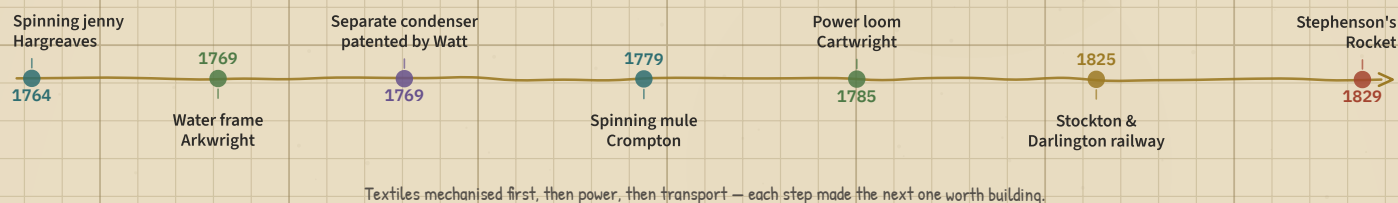
- **Unsorted input.** The precondition is not checked at runtime — the function silently returns the wrong answer.
- $mid = (lo + hi) / 2$ **overflows** for large indices in fixed-width integer types; use $lo + (hi - lo) / 2$.
- Updating $lo \leftarrow mid$ instead of $mid + 1$ leaves the interval unchanged when it has two elements — an **infinite loop**.
- With duplicates, the *first* match is not guaranteed — you need a lower-bound variant.

THE INDUSTRIAL revolution

What changed DEFINITION

The Industrial Revolution — the transition, beginning in Britain in the later eighteenth century, from an economy of **hand production in households and workshops** to one of **powered machinery in factories**. Output, urban population and the use of coal all rose together.

Key inventions



Why Britain first

PRECONDITIONS IN BRITAIN

Resources

- Accessible coal fields
- Iron ore near the coal
- Navigable rivers and canals

Capital and markets

- Colonial and Atlantic trade
- Developed banking and credit
- Growing domestic demand

Institutions

- Patent protection from 1624
- Relatively stable property law
- Enclosure freed rural labour

No single cause is sufficient — historians argue about weighting, not about the list.

Before and after

COTTAGE INDUSTRY

- Work in the home, at the worker's own pace
- Hand tools; the spinner owned the wheel
- Family labour, paid by the piece
- Output limited by hands available



FACTORY SYSTEM

- Work in a mill, to the clock and the machine
- Powered machinery owned by the mill owner
- Waged labour under supervision
- Output limited by power and capital

Consequences

- **Urbanisation** — Manchester grew roughly tenfold across the period.
- **Sanitation lagged** — cholera epidemics from 1831.
- **Child labour** in mills, restricted by the Factory Acts from 1833.
- **Coal consumption** rose steeply as steam replaced water power.

Key takeaway

The decisive change was not any single machine but the **shift of power, capital and discipline out of the household and into the mill** — which is why the **factory system**, not the steam engine, is the unit historians compare across countries.

How the period is dated TECHNICAL NOTE

The dates are a **historians' convention**, not a boundary in the sources.
c. **1760–1840** is the common range; some accounts run to **1850**.

- The phrase was popularised in English by **Arnold Toynbee's** lectures in the 1880s — a century after the events.
- Growth was **uneven by region and by trade**: textiles mechanised early, agriculture and building far later.

Exam point

- Watt did not invent the steam engine — **Newcomen's** engine predates him. Watt's **separate condenser** made it efficient enough to leave the mine.
- Mechanisation of **spinning** came before **weaving**; the resulting yarn surplus is what made the power loom worth building.
- Treat 'the Industrial Revolution' as a **process**, not an event — the dates are conventional, not sharp.